

Engineering Notebooks

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Purpose

- Notebooks are **legal documents** establishing what and when you did something. Dated signatures certify the entries and that you wrote them before their signing.
- Notebooks record what you were doing. They summarise what you have learned and achieved. They allow other people to recreate and continue your work.
- Write down what you do **before** you do it.
- **If you did not write it down, it did not happen.**

Have peers read the notes and give feedback. Do they think information is missing?

Medium

The medium must guarantee the integrity of the notebook. You must not remove, alter, or add anything after writing it. Tampering must be visible.

Electronic notebooks are still uncommon and expensive. Such software must comply with, e.g., EU good manufacturing practice (GMP) Annex 11. Other domains don't allow electronic notes at all. Judges and arbiters prioritise handwritten notebooks over electronic ones. A thread-bound notebook with consecutively numbered pages is usually accepted. It should be at least A5 size. Preferred sizes are A4 and US Letter.

A paper notebook is independent of electricity and cannot be ransom-ware. It still can be destroyed by fire and flood.

Pens

Use a pen that uses archival ink conforming with ISO 12757-2, ISO 14145-2, or ISO 27668-2. Many countries allow fountain pens that use registrar's or iron-gall ink. Check the local legislation.

Pencils and Frixion pens **do not qualify**.

Corrections

Make corrections to see both the original error and the correction. Often, we find the original to have been correct after all, and it is helpful to be able to read it. Draw a **single** thin line through the error. Never use correction fluid or correction tape. Never black-out text. If you correct after entering subsequent text for the same date, note the location of the corrected material next to the struck-out text.

You may correct an entry that you have made on a previous date. The procedure given here is a common approach.

Any correction of an entry from a previous date must be clearly discernible from the entry itself. You accomplish this easily using a **different colour** of ink.

The correction must be **dated and initialled**.

The scope of the correction must be clearly delineated, typically with a bracket around the target entry.

You must make a forward reference to the notebook's page containing an explanation of the correction. You must make an entry explaining the nature and reason for the correction on the date that you applied the correction.

Structure

The notebook has a title page with the dates of entries and your contact details. Company notebooks frequently display the owner's name and the company's contact information.

Leave the first and last two pages empty (they are usually glued to the binding and are fragile).

Leave four pages for a table of contents.

The entries follow the table of contents.

All pages must be numbered consecutively.

Dates

Keep the date format unambiguous. I recommend spelling out the month or using a common abbreviation. Always write the full year. Write "26 August 2025" or "26 Aug 2025".

Labels

Use labels to identify content on a page, like "Figure 15" or "Exhibit A". Make sure labels are unique. Many number the labels consecutively. Reference them liberally: "The setup follows Experiment 8 on page 25".

Exhibits

Add printouts, photographs, or notes taken on paper as exhibits. Leave space for the exhibit by drawing a rectangle and crossing it out. Label it uniquely and give it a title. Once you have the exhibit, glue it to the reserved space. Preferably, glue it so one can check what was below the exhibit.

Sign the exhibit with a paraph and the date of adding it so that the signature is on both the exhibit and the notebook. The supervisor should write a note that they saw the exhibit the next time they witness the notebook.

The Entry

Entries start on a new page, are dated and have a title.

Often, all pages must be dated and have a title. Some open new entries only on the right-facing page (recto). Cross out empty blocks of space.

Leave sufficient space. Especially leaving the spaces left and right of figures, tables, or exhibits empty. Leave at least **3 cm** at the bottom of each page for signatures.

Sign and date each entry at its bottom.

Have a witness, e.g., a teaching assistant or teacher, sign and date the entry. The witness may sign at a later date.

Attendance

Record the names of all attendants when taking meeting minutes or logging group work. Note when people join or leave. You should note the meeting time, have attendants read the minutes, and have them sign.

Give credit in your notes. "We" should only be written if it applies to all attendants. Use "I" or names. Keep names unambiguous. If two attendants have the same name, add a disambiguating property (company name, position, birth year).

Introduction

The introduction contains a brief discussion of the assignment's crucial points. Define its goal: build, implement, test. Start the introduction with the assignment number and title at the top of the sheet. Include any theory, formulas, definitions, designs, and hypotheses related to the assignment.

Answer pre-experimentation questions in the introduction. Please do not repeat the questions, but it should be apparent from the writing which questions you answer. You may place a copy of the assignment just after the introduction.

Log

The goal of the log is to provide enough information to reproduce the work in the future. Details are important, so write everything down, including the machine, the operating system, the software versions, and the parameters.

Record log entries in chronological order. Do not leave space to add content later!

You may enumerate the steps.

It works best if you first write down the step, do it, and then record the results or any other observation. This makes sure that you do not omit crucial observations. All observations are data.

Data

Data is output generated from the computer system or measurements during tests. It may also include summary information, the time it took to execute a program, or the fact that the program crashed.

You can print out data like spreadsheets or log files and enter them as an **exhibit**.

Record **all** data neatly in tables. All tables must be labelled at the top and contain a caption (describing the data recorded) for later reference in your diary, analysis, or summary.

Each column should have a header and include the units for the data in the corresponding column.

Carefully consider whether the recorded data is consistent. Note any outliers or other abnormalities. Strike through erroneous values with a single line and record the correct value. Do not erase or black out wrong values.

Figures

Figures include plots or graphs (visual representations of data), screenshots, or sketches (e.g., for designs to note trends).

Each figure should be labelled and given a caption for later reference in diary entries or analysis.

Plots and graphs must contain labels for each axis with appropriate units. Include a legend with a label for each line if the plot or graph contains lines (e.g., drawing lines through data points).

Source Code

Include the source code for programs written and used to answer questions during the experiment. The source code should be well structured and documented, explaining the program's work. Recording source code is crucial, as others must reproduce the work. If the source code is less than 20 lines, including it in the notebook is adequate. You can add longer code samples as exhibits, as printouts or CD-ROMs. Alternatively, use a version control system like git. Commit the version and reference it in the notebook: "We have used revision 9264e34 of our code at <https://github.com/wi77/LS2.git>."

Analysis

Analyse the results. Include several references to texts and papers in the analysis section. Remember, everything is assumed to be false unless supported by your data and other references.

Discuss the meaning of the data collected. Compare the results to the relevant theoretical background. Note points of agreement and disagreement with the theory. Note any other trends obtained from plots and generate new hypotheses.

Answer the post-experimentation questions posed in the assignment. Identify the question with an introductory sentence and a paragraph answering the question.

Avoid using bullets or numbered lists; their use is only good practice to highlight essential points of the explanation.

Summary

Each lab write-up should include a brief (up to 1 page) summary of the point under study. Summarise the observations and conclusions during the laboratory assignment.

Maintaining the Notebook

We recommend creating a template or boilerplate that contains an outline of the essential notebook sections. It is acceptable to refer to earlier experiments, for example, that the experimental conditions, such as computer platform and software, are the same as those in "Experiment 1: Using Telepathy to Move Mountains".

Use abbreviations as long as they are clearly defined elsewhere in the notebook. When citing other sources, include them using APA style. It is sufficient to cite later citations using the author and year. We do not recommend having a bibliography at the back of the notebook.

Remember, the principal purpose of a notebook is to allow someone else to reproduce the work. All that may be available for this work is what we can read from the notebook. Most people can only read minds when we are around to explain what we wrote.